

DTU



Perception for Autonomous Systems 34759:

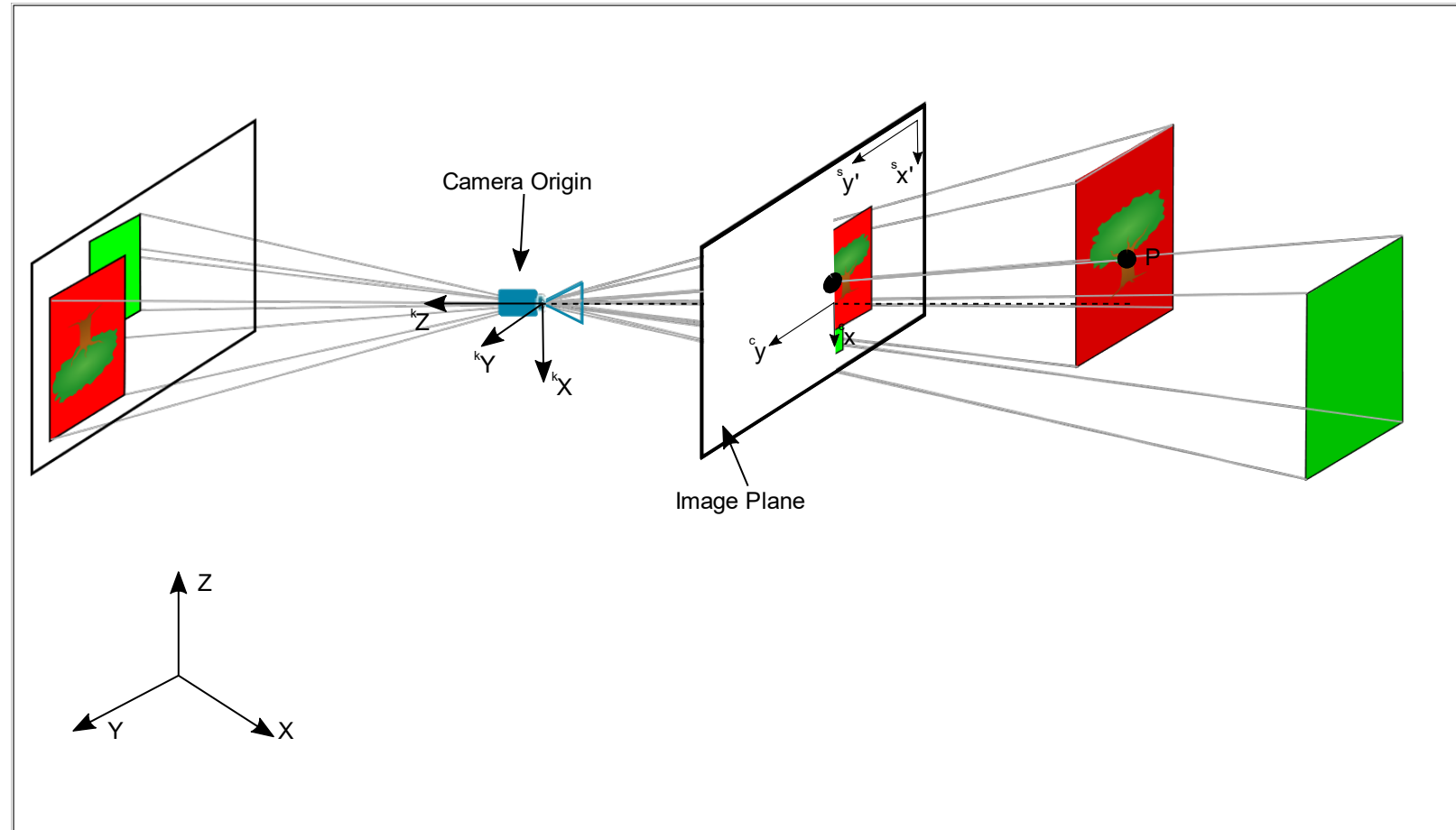
Camera Matrix and Camera Calibration

Lecturer: Evangelos Boukas—PhD

- Image formation
- Camera model – Project from World to sensor
- Camera calibration – The 3D case
- Camera calibration – The realistic case
- Camera radial distortion

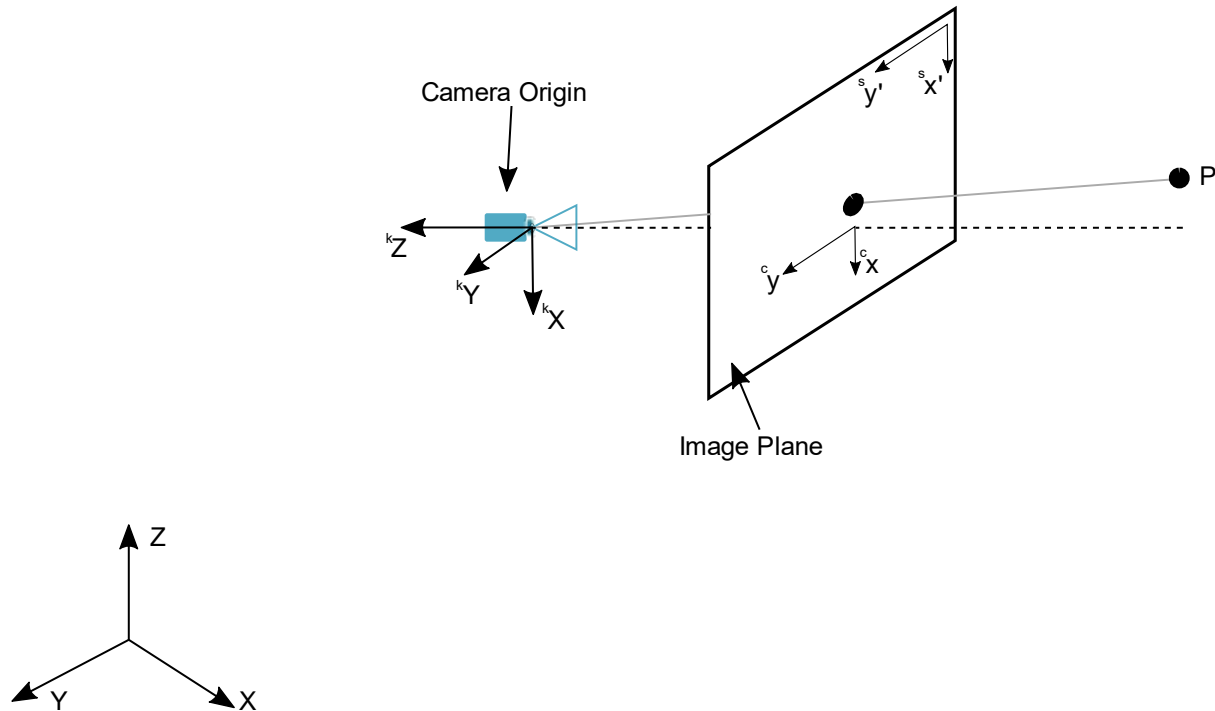
Image Formation

- Let's look at the most simplistic case. What's weird about it?
- Let's Fix that
- Let's get some notation



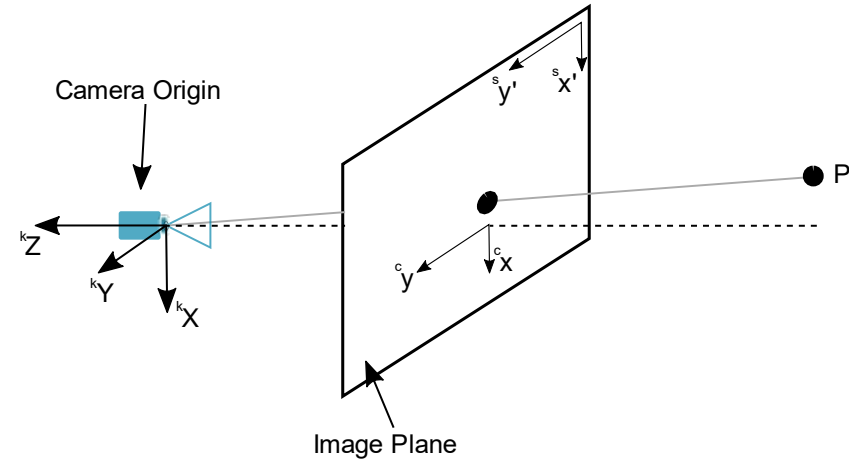
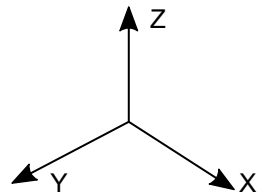
Camera Extrinsics and Intrinsics

- The orientation of the camera wrt World coordinates is called extrinsics
- The rest of Parameters are called intrinsics
- We will see how points in 3D are projected on the sensor



Camera Extrinsics

- We need to express a 3D point P
 $\mathbf{X}_p = [X_p, Y_p, Z_p]^T$
 in the camera coordinates
 ${}^k\mathbf{X}_p = [{}^kX_p, {}^kY_p, {}^kZ_p]$
- To do so we need to use
 the camera Origin
 ${}^k\mathbf{X} = [{}^kX, {}^kY, {}^kZ]$
- Then:
 ${}^k\mathbf{X}_p = [R | T] \mathbf{X}_p = [R | -R^k\mathbf{X}] \mathbf{X}_p$
- $= {}^kR [I | -{}^k\mathbf{X}] \mathbf{X}_p = {}^kR [I | {}^kT] \mathbf{X}_p$



Camera Extrinsics

- So now we have a description for the 3D point in camera frame

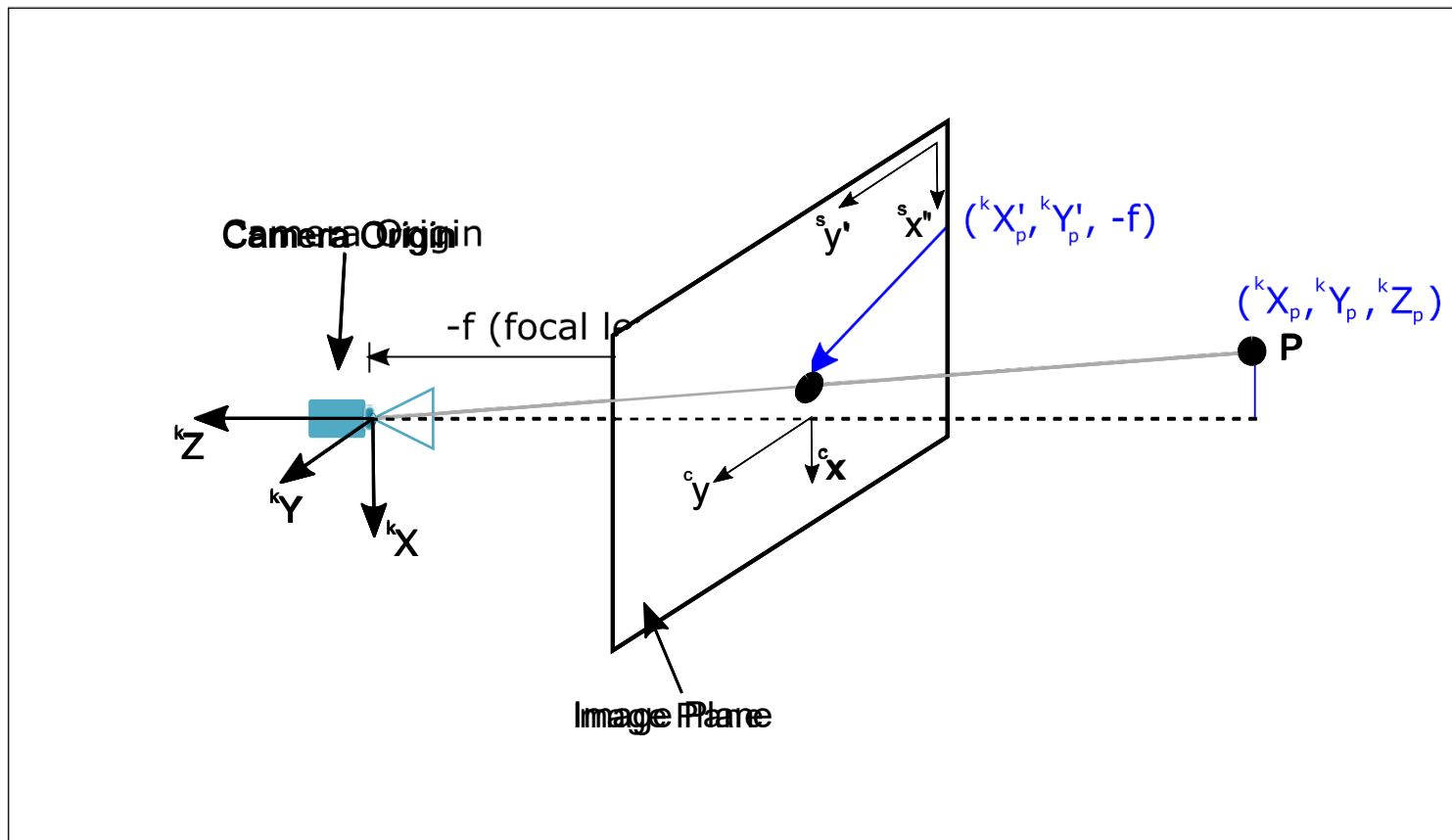
$$[X_p, Y_p, Z_p, 1]^T = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} [X, Y, Z, 1]^T$$

rotation translation Homogeneous Point
In World Coordinates

- To get to the image frame we will have to drop a dimension

DTU Projection

- Let's get rid of things we don't need
- So it is clear that the position of the projection on the image frame is dependent on the ratio X/Z
- In fact ${}^c x = -f * {}^k X_p / {}^k Z_p$
and ${}^c y = -f * {}^k Y_p / {}^k Z_p$
- Additionally we can add the transformation to the sensor frame



Projection

- From the previous

$$[X_p, Y_p, Z_p, 1]^T = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} [X, Y, Z, 1]^T$$

rotation translation Homogeneous Point
In World Coordinates

- We get to the following

$$[{}^sX, {}^sY, W]^T = \begin{bmatrix} -f s_x & 0 & x'_c \\ 0 & -f s_y & y'_c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} [X, Y, Z, 1]^T$$

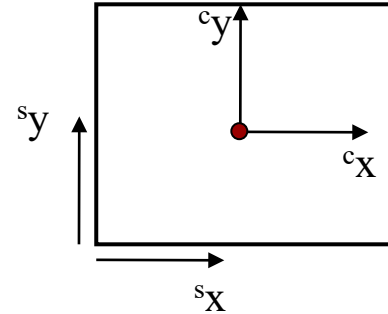
Intrinsics rotation translation Homogeneous Point
In World Coordinates

- Where W is the homogenous scale
- This is the analytic version of the projection matrix**

Projection Matrix

- We can define the Projection Matrix as follows

$$\begin{bmatrix} \lambda x \\ \lambda y \\ \lambda \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



- **This is great news for Calibration!**
- It means that our problem is a system of linear equations
- How many unknowns?
- 11: (5 + 6) Intrinsic+Extrinsic

Calibration – Known 3D positions

- We defined our camera projection as follows:

$$\begin{bmatrix} \lambda x \\ \lambda y \\ \lambda \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- If we “sample” a set of points in 3D and 2D: $[X, Y, Z] \rightarrow [x, y]$
we get the following analytic equations:

$$\lambda x = P_{11}X + P_{12}Y + P_{13}Z + P_{14}$$

$$\lambda y = P_{21}X + P_{22}Y + P_{23}Z + P_{24}$$

$$\lambda = P_{31}X + P_{32}Y + P_{33}Z + P_{34}$$

- However λ is used for scale and therefore does not provide a 3rd equation

Calibration – Known 3D positions

- So for every 3D to 2D correspondence we get 2 equations:

$$(P_{31}X + P_{32}Y + P_{33}Z + P_{34})x = P_{11}X + P_{12}Y + P_{13}Z + P_{14}$$

$$(P_{31}X + P_{32}Y + P_{33}Z + P_{34})y = P_{21}X + P_{22}Y + P_{23}Z + P_{24}$$

- How many points do we need?
- 6 Points x 2 equations

- For every point we get the following

$$0 = p_{11}X + p_{12}Y + p_{13}Z + p - p_{31}xX - p_{32}xY - p_{33}xZ - p_{34}x$$

$$0 = p_{21}X + p_{22}Y + p_{23}Z + p_{24} - p_{31}yX - p_{32}yY - p_{33}yZ - p_{34}y$$

- We can format all the linear equations in a single matrix: $Ap=0$

$$\begin{bmatrix} X_1 & Y_1 & Z_1 & 1 & 0 & 0 & 0 & 0 & -x_1X_1 & -x_1Y_1 & -x_1Z_1 & -x_1 \\ 0 & 0 & 0 & 0 & X_1 & Y_1 & Z_1 & 1 & -y_1X_1 & -y_1Y_1 & -y_1Z_1 & -y_1 \\ & & & & & & \vdots & & & & & \\ X_n & Y_n & Z_n & 1 & 0 & 0 & 0 & 0 & -x_nX_n & -x_nY_n & -x_nZ_n & -x_n \\ 0 & 0 & 0 & 0 & X_n & Y_n & Z_n & 1 & -y_nX_n & -y_nY_n & -x_nZ_n & -x_n \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ p_{14} \\ p_{21} \\ p_{22} \\ p_{23} \\ p_{24} \\ p_{31} \\ p_{32} \\ p_{33} \\ p_{34} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Calibration – Known 3D positions

- The solution is given by the following process which is itself called DLT:

- Calculate the Singular Value decomposition (SVD)

$$\text{SVD} : A = UDV^T$$

- *Get the last column of V*

$$P = V_{\text{smallest}} \text{ (column of } V \text{ corr. to smallest singular value)}$$

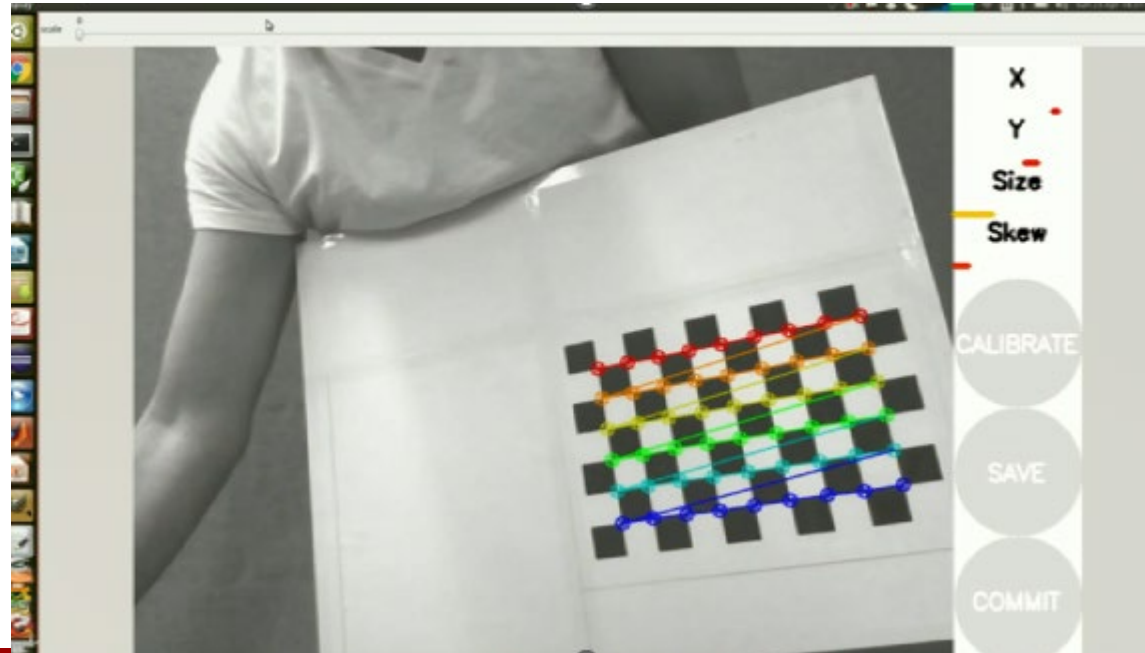
- Reshape into the P matrix

Calibration – Known 3D positions

- To summarize:
 - Get min 6 3D – 2D correspondences
 - Form matrix A ($AP=0$)
 - Calculate SVD
 - Get Last column of V which corresponds to the values of P

Calibration in real world

- In Real World we do not have 3D points.
- Or do we?
- What if we could use some sort of predefined setup which would allow us to know the relative position of 3D points?
- It turns out that if we have a calibration pattern we can know their relative positions of all points as they are on a known configuration





Calibration using homography

- Since $Z = 0$ we lose one degree of freedom:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}$$

- We can then redefine our P as:

$$\equiv \begin{bmatrix} P_{11} & P_{12} & P_{14} \\ P_{21} & P_{22} & P_{24} \\ P_{31} & P_{32} & P_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

- This is called homography and allows us to project points from one plane to another

Calibration using homography

- How to calculate the homography:

$$\mathbf{x}' = \mathbf{H}\mathbf{x} \quad \mathbf{x}' = \begin{bmatrix} w'x' \\ w'y' \\ w' \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix}$$

- The homography has 9 elements. However, similar to before due to the scale there are only 8 unknowns
- Therefore we need minimum 4 correspondences to setup the problem as a DLT

$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & xy' & x' \\ 0 & 0 & 0 & -x & -y & -1 & yx' & yy' & y' \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \\ h_9 \end{bmatrix} = \mathbf{0}$$

Calibration using homography

Steps

- With a static camera move the pattern and get images.
 - Make sure to not present only one orientation, as there will be a problem with the estimation of the homography
- Calculate correspondences using the pattern
- Solve using **Normalized DLT**
 - Normalize coordinates for each image
 - Translate for zero mean
 - Scale so that average distance to origin is $\sim\sqrt{2}$
 - Form Matrix A
 - Solve using SVD
 - Calculate SVD
 - Use last column of V to get the homography.
 - Denormalize

Calibration using homography

- However the DLT solution is known to be prone to outliers.
- Other approaches exist to solving the calibration:
 - Nonlinear least squares
 - RANSAC
- RANSAC:
 1. Choose number of samples N
 2. Choose 4 random potential matches
 3. Compute $\mathbf{H}$ using normalized DLT
 4. Project points from $\mathbf{x}$ to $\mathbf{x}'$ for each potentially matching pair:
 5. Count points with projected distance $< t$
 6. Repeat steps 2-5 N times

Choose $\mathbf{H}$ with most inliers

Calibration - Distortion

- Ok, but still What about the lens Distortion?



Barrel



Pincushion



Fisheye

Calibration - Distortion

- So how do we model it:
- Usually a quartic (biquadratic) polynomial is used:

$$\hat{x}_c = x_c(1 + \kappa_1 r_c^2 + \kappa_2 r_c^4)$$

$$\hat{y}_c = y_c(1 + \kappa_1 r_c^2 + \kappa_2 r_c^4)$$

- OR higher order

Calibration - Distortion

- So how do we model it:
- Usually a quartic (biquadratic) polynomial is used:

- OR h



Distorted

+

+



Undistorted

Calibration - Distortion

- How to calibrate for the distortion coefficients:
 - First do DLT for the Projection Matrix
 - Then form a non linear least square problem that includes both the linear projection and the non-linear distortion using the Levenberg Marquardt algorithm.



Conclusion

- What did we learn?

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